Module 4.1: Statistical Reasoning - Key Terms & Definitions, PART 1

Statistical Reasoning – Key Terms & Definitions, PART 1

The **Bar Chart**: is a graphical display for visualizing the **distribution** of **categorical** data. It emphasizes *how the different categories compare with each other*.

Data: are pieces of information about ***individuals (a particular person)*** or ***objects*** organized into **variables**.

Dataset: is a set of data identified with particular circumstances. Datasets are typically displayed in tables, in which *rows* represent ***individuals*** and *columns* represent ***variables***.

Deviation: measures how far individual data points differ from a central value, usually the **mean**.

Distribution: is *a function that shows the possible values for a variable* and how often they occur. In other words, by **distribution** of a **variable**, we mean:

- what value the variable takes, and
- how often the variable takes those values.

Exploratory Data Analysis (EDA): involves *using graphics and visualizations to explore and analyze a data set*. The goal is to explore, investigate, and learn about the collected data.

- When performing **EDA**, we should always:
    - use *visual displays* (graphs or tables) plus *numerical summaries.*
    - describe the *overall pattern* and mention *striking deviations* from that pattern.
    - interpret the results we got *in context*.

Histogram: is a graphical display for visualizing the **distribution** of a **quantitative variable**.

- Idea: Break the range of values into intervals and count how many observations fall into each interval (see example C in Module 4.2.4, pp. 14-16).
- Once the **distribution** has been displayed graphically, we can describe **the overall pattern of the distribution** and mention any striking **deviations** from that pattern. More specifically, we can consider the following features of the **distribution**:

- Shape
- Center
- Spread

overall pattern

- Outliers → deviations from the pattern

Inference: To draw a conclusion from the **sample** about the **population**.

The **Pie Chart:** is a graphical display for visualizing the **distribution** of **categorical** data. It emphasizes *how the different categories relate to the whole.* **Pie charts** can be useful for giving a high-level overview to show how a set of cases breaks down.

Mean ($\bar{x}$): is the average of a set of observations (i.e., the sum of the observations divided by the number of observations).

$$\text{mean} = \frac{\text{sum of values}}{\text{number of values}}$$

If the *n* observations are $x_1, x_2, \ldots, x_n$, their **mean**, which we denote by **$\bar{x}$ (and read 'x-bar'**), is therefore

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

- **Example:**
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Example **(H) Best Actress Oscar Winners**

Again we use the Best Actress Oscar winners example.

34 34 27 37 42 41 36 32 41 33 31 74 33 49 38 61 21 41 26 80 42 29 33 36 45 49 39 34 26 25 33 35 35 28
29 61 32 33 45 29 62 22 44

The mean age of the 32 actresses is

$$\bar{x} = \frac{34 + 34 + 27 + \cdots + 62 + 22 + 44}{44} = \frac{1687}{44} = 38.3.$$

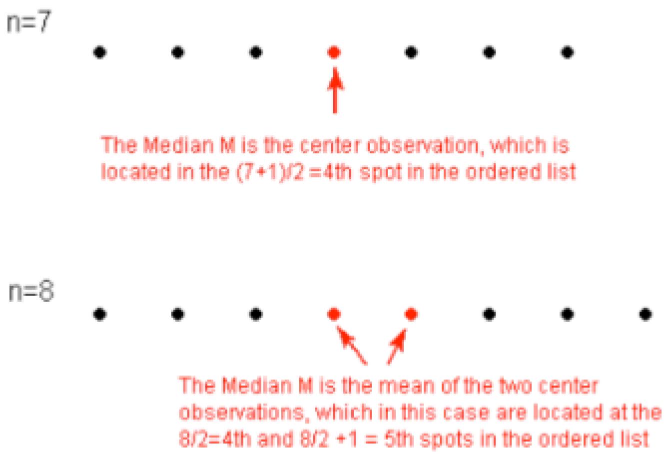
- The **mode** is the most frequently occurring value. The **mean** is the average value, while the **median** is the middle value.

Median (M): is the midpoint of the **distribution** (splits the data in half). It is the number such that half of the observations fall above, and half fall below.

- **IMPORTANT:** To find the **median:**
- Order the data from smallest to largest.
- Consider, whether **n**, the number of observations, is **even** or
  - – If n is odd, the median *M* is the center observation in the ordered list. This observation is the one “sitting”in the (n + 1)/2 spot in the ordered list.
  - – If n is even, the median *M* is the **mean** of the two center observations in the ordered list. These two observations are the ones “sitting” in the n/2 and n/2 + 1 spots in the ordered list.
    - Put differently, if there are an even number of observations, there will be two values in the middle, and the **median** is taken as their average.

Example (J) Median (1)

For a simple visualization of the location of the median, consider the following two simple cas
8 ordered observations, with each observation represented by a solid circle:



Example (K) Median (2)

To find the median age of the Best Actress Oscar winners, we first need to order the data. It would be useful, then, to use the stemplot, a diagram in which the data are already ordered.

Here n = 44 (an even number), so the median M, will be the mean of the two center observations. These are located at the n / 2 = 44 / 2 = *22nd* and n / 2 + 1 = 44 / 2 + 1 = *23rd* spots. Counting from the top, we find that:

- the 22nd ranked observation is 34
- the 23rd ranked observation is 35

Therefore, the median M = (34 + 35)2 = 34.5

```
2|12
2|56678999
3|0122333333344
3|5566789
4|1112244
4|599
5|
5|
6|112
6|
7|4
7|
8|0
```

- The **mode** is the most frequently occurring value. The **mean** is the average value, while the **median** is the middle value.

Mode: is the most commonly occurirng value in a distribution. It is represented by a prominent **peak** in the **distribution**.

- **Example:**

Example (F) Best Actress Oscar Winners

We will continue with the Best Actress Oscar winners example.

To find the most commonly occurring, or *modal*, age, it is helpful to list the ages in a frequen
the following results:

Best Actress Age	21	22	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	41	42	44
Count	1	1	1	2	1	1	3	1	1	2	6	2	2	2	1	1	1	3	2	2

The mode is 33, since it occurs the most times (6).

- The **mode** is the most frequently occurring value. The **mean** is the average value, while the **median** is the middle value.

Modality: Peakedness. The number of peaks (**modes**) the **distribution** has. We can distinguish between **unimodal (one mode)** **and** **bimodal (two modes)**.

Outliers: are data points that fall outside the overall pattern of the **distribution**.

Peakedness: Modality. The number of peaks (**modes**) the **distribution** has.

Pictogram: A variation on **pie chart** and **bar chart**.

Population: The whole collection (entire group) of individuals under the study. Note that the word **population** here does not refer only to people but also to animals, objects, etc.

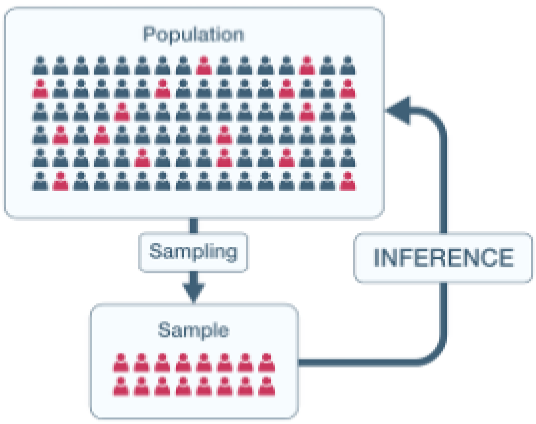
- Example:
  - The opinions of the population of U.S. adults about the death penalty.
  - How the population of mice reacts to a certain chemical.
  - The average price of the population of all one-bedroom apartments in the entire city.

Probability: is the "machinery" (of the 'Big Picture of Statistics') that allows us to draw conclusions about the **population** based on the data collected about the **sample**.

Producing Data: Choosing a **sample** from the **population** and collecting data from it.

Sample: A subgroup of the **population** / the observed members of the target group (**population**) from which data are collected.

- Example:



Scale of Measurement: A precise method to categorize **variables**

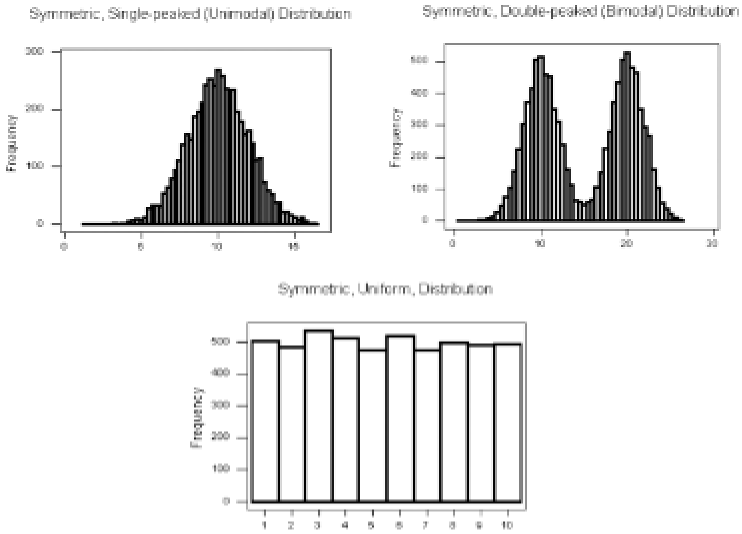
- There are four different scales of measurement:
    1. **Nominal Scale of Measurement:** is a qualitative measure that uses discrete categories to describe a characteristic of the research participants.
    2. **Ordinal Scale of Measurement:** rank-orders participants on some scale or attribute, but the difference between numbers does not convey fixed or equal differences.
    3. **Interval Scale of Measurement:** takes numerical form, and the distance between pairs of consecutive numbers is assumed to be equal.
    4. **Ratio Scale of Measurement:** is similar to the **interval scale** – as with the interval scale, a number is assigned to a subject that represents the amount of the attribute that the subject has and the difference between consecutive numbers is assumed to be equal.

See also: https://www.youtube.com/watch?v=dbI8EDP8IxE

Shape of a distribution: describes how frequently values occur. Here, we should consider

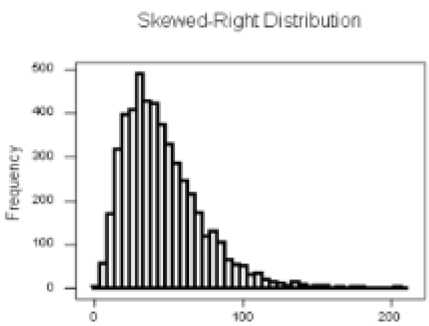
1. **Symmetry/Skewness** of the **distribution**
2. **Peakedness (modality)** – the number of **peaks (modes)** the distribution has.

1. **Symmetric Distribution**



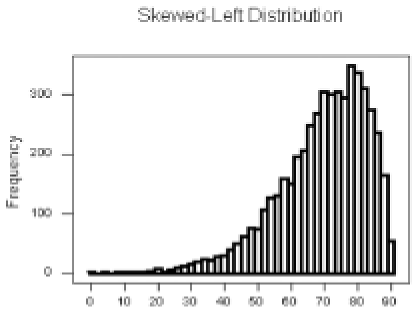
All three **distributions** are **symmetric**, but are different in the **modality (peakedness)**. The **first distribution** is **unimodal** – it has **one mode** around which the observations are concentrated. The **second distribution** is **bimodal** – it has **two modes** around which the observations are concentrated. The **last distribution** is **flat (uniform) – and has no modes, or no values around which the observations are concentrated.**

2. Skewness: Using skewness, we can distinguish between **skewed right distributions** and **skewed left distributions**:



Skewed Right Distribution: The right tail (larger values) is much longer than the left tail (small values). (When data trail has a longer right tail, the shape is said to be **skewed right**).

- Other ways to describe data that are skewed right: skewed to the right, skewed to the high end, or skewed to the positive end.



Skewed Left Distribution: The left tail (smaller values) is much longer than the right tail (larger values). (Data sets with the reverse characteristic – a long, thinner tail to the left – are said to be **skewed left**).

- **IMPORTANT:** An example of a real life variable that has a **skewed left distribution** is age of death from natural causes (heart disease, cancer, etc.). Most such deaths happens at older ages, with fewer cases happening at younger ages.

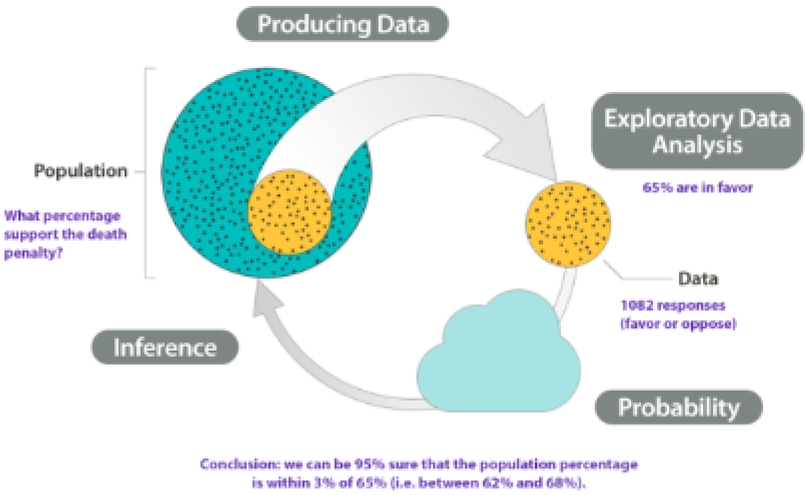
Statistics: converts raw data into useful information through 1) collection (**producing data**), 2) summarizing (**EDA**), and 3) interpretation of data (**probability & inference**):

At the end of April 2005, a poll was conducted (by ABC News and the *Washington Post*) for the purpose of learning the opinions of U.S. adults about the death penalty.

1. *Producing Data:* A (representative) sample of 1,082 U.S. adults was chosen, and each adult was asked whether he or she favored or opposed the death penalty.

2. *Exploratory Data Analysis (EDA):* The collected data were summarized, and it was found that 65% of the sampled adults favor the death penalty for persons convicted of murder.

3 and 4. Probability and Inference: Based on the sample result (of 65% favoring the death penalty) and our knowledge of probability, it was concluded (with 95% confidence) that the percentage of those who favor the death penalty in the population is within 3% of what was obtained in the sample (i.e., between 62% and 68%). The following figure summarizes the example:



Stemplot: (also called a stem or leaf plot) is a graphical display of the **distribution** of **quantitative** data. The **stemplot** has additional unique features: it is easy and quick to construct for small, simple datasets; it preserves the original data; and it sorts the data.

Spread: describes the *variation* of the data. Three most commonly used measures of spread are **range, inter-quartile range (IQR)**, and **standard deviation (SD)**. (See Statistical Reasoning - Key Terms & Definitions, PART 2)

Variable: A particular characteristic of the ***individual***.

- **Variables** can be classified into *one of two types*:

- **Quantitative variables:** take *numerical values* and *represent some kind of measurement* (e.g., weight, age, height, etc.)
- **Categorical variables:** take category or label values and place an individual into one of several groups. Each observation can be placed in *only one category*, and the *categories are mutually exclusive* (e.g., zipcode, gender, race, smoking yes/no, etc.)
    - The **distribution** of a **categorical variable** is summarized using *graphical display:* **pie chart** or **bar chart**, supplemented by *numerical summaries*: category counts and percentages.
        - **Example!**

Example (C) Exam Grades

Here are the exam grades of 15 students:

Exam Grades

Score	Count
[40–50)	1
[50–60)	2
[60–70)	4
[70–80)	5
[80–90)	2
[90–100]	1

Note: The observation 60 was counted in the 60–70 interval. See comment 1 below.

To construct the histogram from this table we plot the intervals on the X-axis, and show the number of obser tions in each interval (frequency of the interval) on the Y-axis, which is represented by the height of a rectar located above the interval:

The table above can also be turned into a relative frequency table using the following steps:

1. Add a row on the bottom and include the total number of observations in the dataset that are represen in the table.
2. Add a column, at the end of the table, and calculate the relative frequency for each interval, by dividing number of observations in each row by the total number of observations.

These two steps are illustrated in red in the following frequency distribution table:

Score	Count (also called Frequency)	Relative Frequency
[40–50)	1	0.07
[50–60)	2	0.13
[60–70)	4	0.27
[70–80)	5	0.33
[80–90)	2	0.13
[90–100]	1	0.07
Total	15	

Step 2: Add a column to right side of table. Determine the relative frequenc of each interval by dividing the interval count (or frequency) by the total numb of observations.

For example, to determine the relative frequency of scores in the [40-50) interval, divide the count (or frequency by the total number of observations: 1 /15 = .07.

The relative frequency for the [50-60) interval is: 2/15 = .13.

Continue until all of the relative frequencies have been calculated.

To convert each relative frequency into percentage, multiply it by 100. For example, the percentage of scores for t [40-50) interval would be .07*100= 7, which is 7%.

Step 1: Add a row at bottom of table. Put in total number of observations in the data set.

In this example, there are 15 (1+2+4+5+2+1= 15) total observations.

1. What is the percentage of exam scores that were 70 and up to, but not including, 80? To determi answer, we look at the relative frequency associated with the [70-80) interval. The relative freque 0.33; to convert to percentage, multiply by 100 (0.33 × 100= 33) or 33%.
2. What is the percentage of exam scores that are at least 70? To determine the answer, we need to:
   - Add together the relative frequencies for the intervals that have scores of at least 70 or above. would need to add together the relative frequencies from [70-80), [80-90), and [90-100] = 0.33+.1 = 0.53.
   - To get the percentage, need to multiple the calculated relative frequency by 100. In this case, it be 0.53 × 100 = 53 or 53%.